Towards Robust Semantic Correspondence: A Benchmark and Insights

Supplementary Material

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A Dataset Construction Details

Building upon the clean subset, we further augment it with four key dimensions of adverse conditions: geometric variations, blur and noise, digital artifacts, and environmental changes. We follow prior works [1] [2] to construct our augmented dataset ensuring comprehensive evaluation. First, we introduce geometric variations, which are constructed as follows: geometric variations include random rotations with angles ranging from 0° to 360°, random perspective warping with control point perturbations to simulate viewpoint changes, and isotropic scaling with factors ranging from 0.1 to 0.9. Blur and noise, as well as digital artifacts, are constructed following the code from ImageNet-C. Defocus Blur applies a circular blur kernel via convolution, mimicking out-of-focus optics. Motion Blur uses a 1D Gaussian kernel along a random angle, simulating camera motion. Gaussian Noise adds zero-mean noise to the normalized image, clipped to [0, 255], replicating random interference. JPEG Compression converts the image to PIL format, saves it as JPEG using BytesIO to simulate the compression process and reduces image detail. Pixelation downsamples the image and upsamples it using nearest-neighbor interpolation, creating a blocky effect. Contrast adjusts pixel values with $(x - \mu) \cdot c + \mu$, lowering contrast. In the environmental changes dimension, images under different conditions are generated as follows: darkness is simulated using gamma correction; fog effects are synthesized by applying an atmospheric scattering model with depth maps and transmission rates; rain effects are created by blending the original image with resized rain-streak images, applying Gaussian blur to the streaks; and snow effects are produced by blending the original image with randomly selected snow mask images through weighted synthesis.

B Experiments Details

B.1 Experimental Set-up

Robustness Evaluation: All semantic correspondence methods are implemented using their official GitHub repositories with default pretrained models. And all experiments are conducted on an NVIDIA GeForce RTX 3090 GPU to ensure consistency in runtime performance. To ensure the validity of our implementation, we first reproduce baseline results on the original SPair-71K dataset, confirming consistency with the performance reported in the respective original papers. We then evaluate all methods on SCAS dataset proposed in this paper. To ensure fair comparisons, we use all methods without any hyperparameter tuning or architectural modifications, evaluating them under identical testing conditions.

Robustness Enhancement: To investigate robustness enhancement, we adopt an experimental setup as [5] and employ the Stable Diffusion v1-5 model as the feature extractor. The default number of time steps for the model is set to $t = 100$, and it receives inputs at a resolution of 960×960. The model is optimized using AdamW [3] with a weight decay of 0.001. We adopt the one-cycle learning rate scheduler [4], with a peak learning rate of 1.25×10^{-3} and 30% of the cycle dedicated to the warm-up phase. All experiments are conducted on a single NVIDIA RTX 3090 GPU.

C Additional Analyses

C.1 Accuracy of Semantic Correspondence Methods

Figure 1 presents the performance of various semantic correspondence methods across four scenarios: Geometry, Blur&Noise, Digital, and Environment. Across all methods, a consistent performance decline is observed when moving from clean to adverse conditions within each scenario. The decline is most pronounced in the Geometry scenario, particularly under the rotation setting, where performance has decreased by an average of 40%. Similarly, the viewpoint setting exhibits a notable drop, with an average decrease of 28.20%. Additionally, significant performance reductions are observed in the gaussian noise scenario (average decline of 16.81%) and in the haze scenario (average decline of 17.22%). In contrast, the decline is less pronounced in the Environment scenario, where methods exhibit minimal performance degradation in snow, rain, and strong sunlight settings. Specifically, accuracy drops by only 1.59%, 0.08%, and 1.37% respectively, indicating that the methods maintain relatively stable performance under these conditions.

C.2 Absolute and Relative Robustness Analysis

In most scenarios, the LVM-based fine-tuning method achieves higher absolute robustness than other approaches, particularly in the scale setting of the Geometry scenario (Figure 2) and the defocus blur setting of the Blur & Noise scenario (Figure 3). In contrast, the LVM-based unsupervised method generally exhibits greater relative robustness. Exceptions are limited to a few settings, such as contrast setting in the Digital scenario (Figure 4) and haze setting in the Environment scenario (Figure 5), where the fine-tuning method outperforms in both absolute and relative robustness. In contrast, the supervised approach consistently performs worse in both absolute and relative robustness compared to the other methods.

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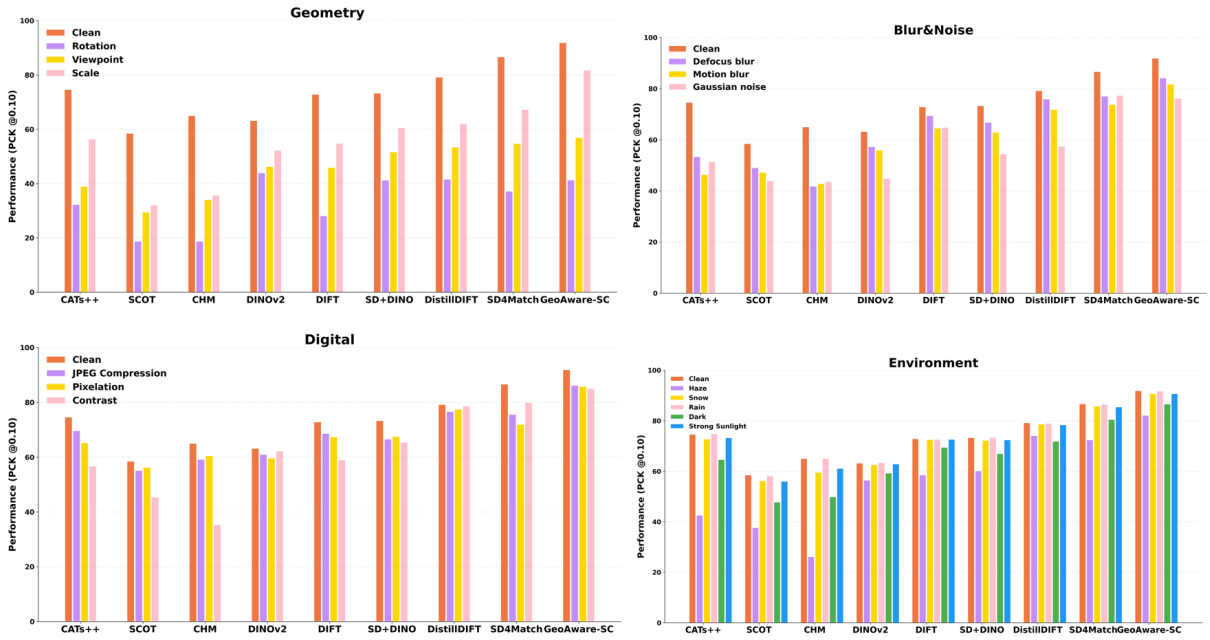

Figure 1: Accuracy of various semantic association methods across four scenarios. The figure consists of four bar charts, each representing the performance of different methods in a specific scenario.

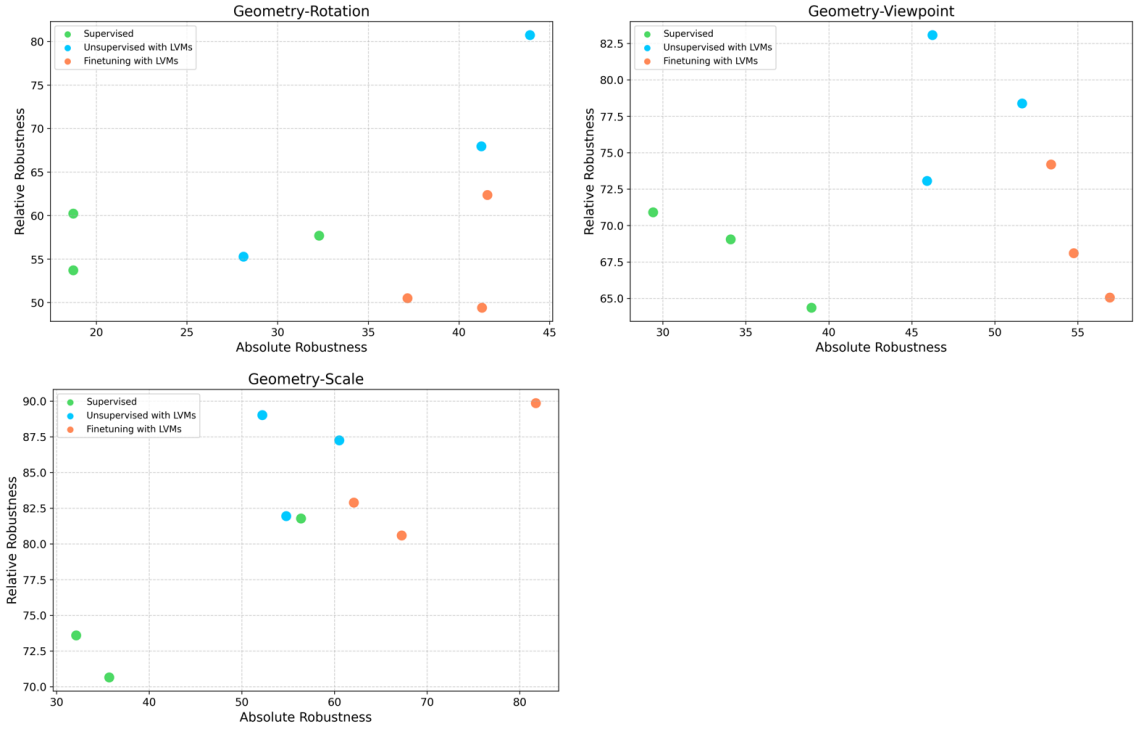

Figure 2: The scatter plot demonstrates the relative and absolute robustness of three categories of methods when subjected to geometric transformations, including rotation, viewpoint changes, and scale variations.

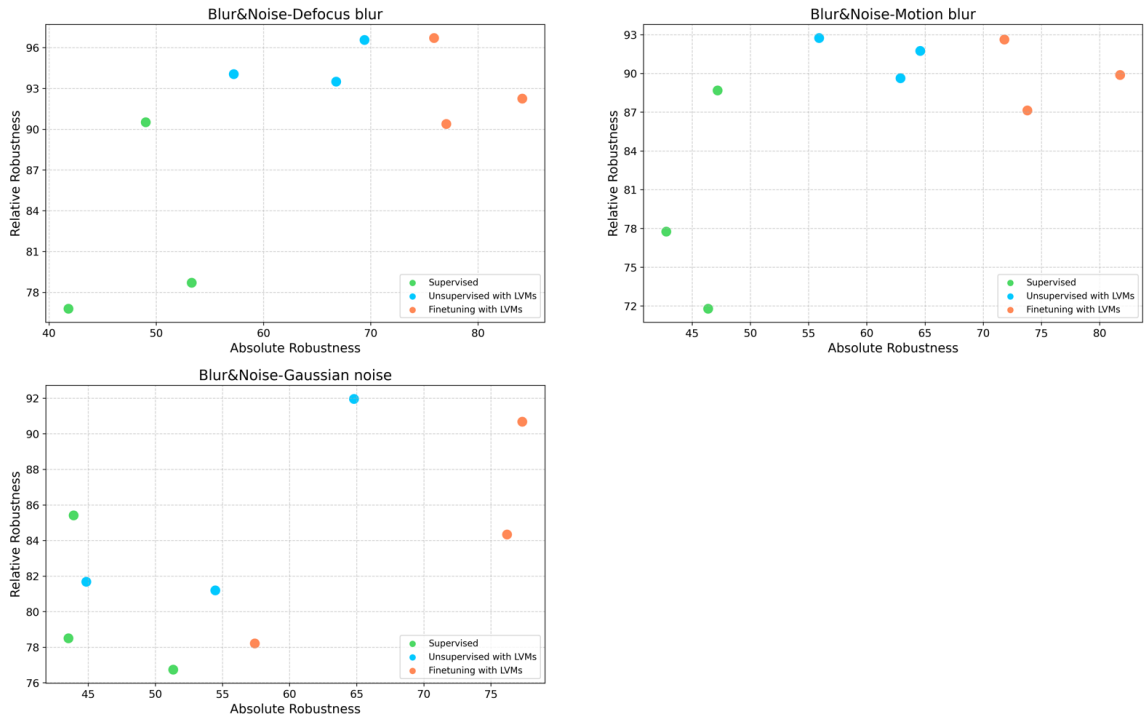

Figure 3: The scatter plot demonstrates the relative and absolute robustness of three categories of methods when subjected to blur and noise transformations, including defocus blur, motion blur, and gaussian noise.

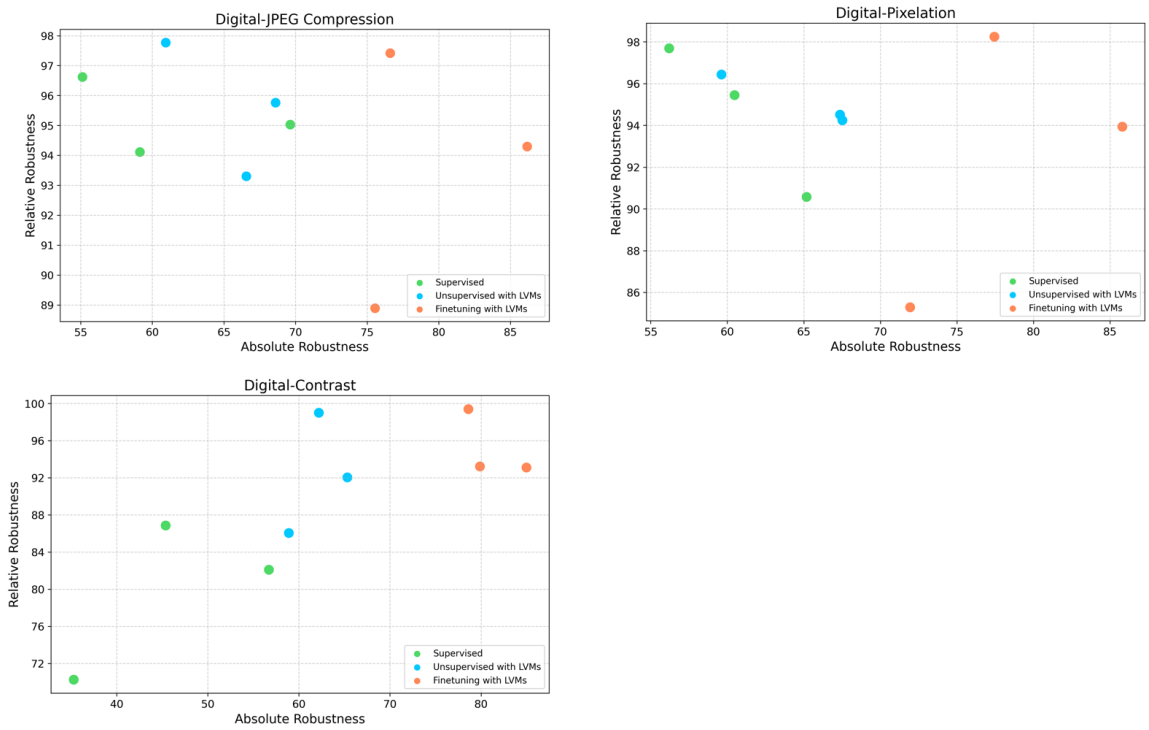

Figure 4: The scatter plot demonstrates the relative and absolute robustness of three categories of methods when subjected to digital transformations, including JPEG compression, pixelation, and contrast changes.

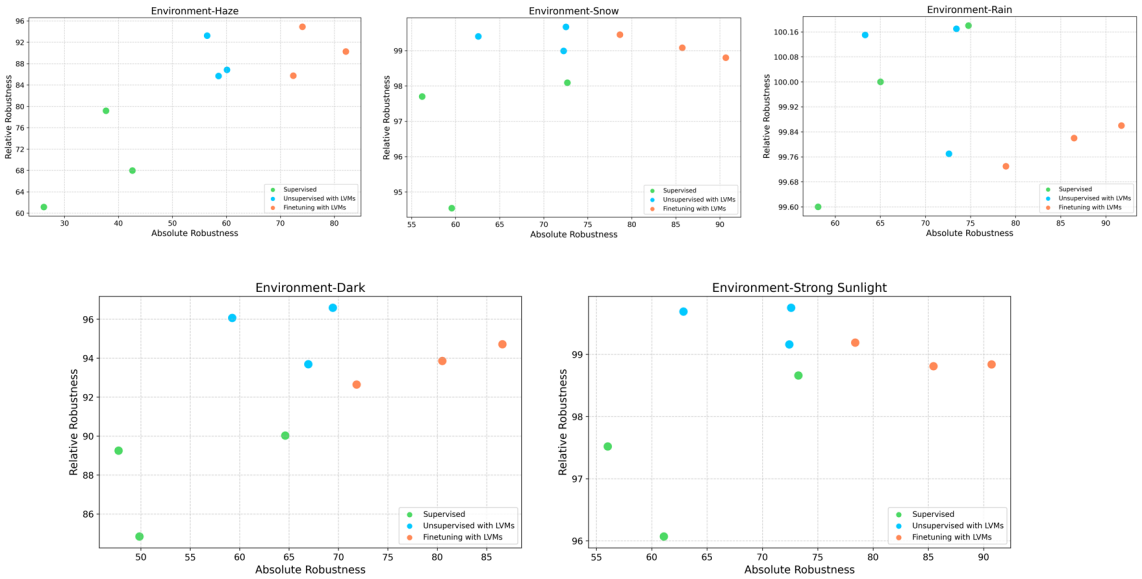

Figure 5: The scatter plot demonstrates the relative and absolute robustness of three categories of methods when subjected to environmental transformations, including haze, snow, rain, darkness, and strong sunlight.

Table 1: Comparison of absolute and relative robustness for DINO, Stable Diffusion, and Stable Diffusion+DINO methods under geometric transformations, including rotation, viewpoint, and scale.

Category	Method	Clean	Absolute Robustness				Relative Robustness			
			Rotation	Viewpoint	Scale	Avg	Rotation	Viewpoint	Scale	Avg
D	DINOv1	46.10	18.72	25.62	28.80	24.38	72.62	79.52	82.70	78.28
	DINOv2	63.17	43.91	46.24	52.19	47.45	80.74	83.07	89.02	84.28
G	SDv1-5	68.71	28.76	42.80	51.55	41.04	60.05	74.09	82.84	72.33
	DIFT->SDv2-1	72.84	28.11	45.91	54.79	42.94	55.27	73.07	81.95	70.10
D&G	SDv1-5+DINOv2	73.26	41.23	51.64	60.51	51.13	67.97	78.38	87.25	77.87

Table 2: Comparison of absolute and relative robustness for DINO, Stable Diffusion, and Stable Diffusion+DINO methods under Blur&Noise conditions, including defocus blur, motion blur, and Gaussian noise.

Category	Method	Clean	Absolute Robustness				Relative Robustness			
			Defocus blur	Motion blur	Gaussian noise	Avg	Defocus blur	Motion blur	Gaussian noise	Avg
D	DINOv1	46.10	38.32	36.06	37.20	37.19	92.22	89.96	91.10	91.09
	DINOv2	63.17	57.23	55.91	44.85	52.66	94.06	92.74	81.68	89.49
G	SDv1-5	68.71	57.23	49.18	48.25	51.55	88.52	80.47	79.54	82.84
	DIFT->SDv2-1	72.84	69.41	64.59	64.80	66.27	96.57	91.75	91.96	93.43
D&G	SDv1-5+DINOv2	73.26	66.76	62.89	54.46	61.37	93.50	89.63	81.20	88.11

C.3 Robustness of DINO vs Stable Diffusion

Methods incorporating Stable Diffusion exhibit superior absolute robustness across fourteen challenging conditions. However, in scenarios involving geometric transformations (Table 1), particularly rotation, DINOv2 achieves the highest accuracy, outperforming other approaches. Furthermore, the DINO series demonstrates greater relative robustness across these conditions. However, under degradation scenarios such as Blur & Noise (Table 2), especially defocus blur and gaussian noise, DIFT achieves the best performance. Meanwhile, under adverse environmental conditions (Table 4) including snow, rain, and strong sunlight, methods leveraging Stable Diffusion consistently outperform other methods.

References

[1] Dan Hendrycks and Thomas Dietterich. 2019. Benchmarking Neural Network Robustness to Common Corruptions and Perturbations. *Proceedings of the International Conference on Learning Representations* (2019).

[2] Yuan Liu, Zehong Shen, Zhixuan Lin, Sida Peng, Hujun Bao, and Xiaowei Zhou. 2019. Gift: Learning transformation-invariant dense visual descriptors via group cnns. *Advances in Neural Information Processing Systems* 32 (2019).

[3] Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).

[4] Leslie N Smith and Nicholay Topin. 2019. Super-convergence: Very fast training of neural networks using large learning rates. In *Artificial intelligence and machine learning for multi-domain operations applications*, Vol. 11006. SPIE, 369–386.

[5] Junyi Zhang, Charles Herrmann, Junhwa Hur, Luisa Polania Cabrera, Varun Jampani, Deqing Sun, and Ming-Hsuan Yang. 2023. A Tale of Two Features: Stable Diffusion Complements DINO for Zero-Shot Semantic Correspondence. *arXiv preprint arxiv:2305.15347* (2023).

Table 3: Comparison of absolute and relative robustness for DINO, Stable Diffusion, and Stable Diffusion+DINO methods under digital conditions, including JPEG compression, pixelation, and contrast.

Category	Method	Clean	Absolute Robustness				Relative Robustness			
			JPEG Compression	Pixelation	Contrast	Avg	JPEG Compression	Pixelation	Contrast	Avg
D	DINOv1	46.10	43.03	43.71	34.60	40.45	96.93	97.61	88.50	94.35
	DINOv2	63.17	60.94	59.61	62.18	60.91	97.77	96.44	99.01	97.74
G	SDv1-5	68.71	53.47	57.46	38.92	49.95	84.76	88.75	70.21	81.24
	DIFT->SDv2-1	72.84	68.60	67.36	58.89	64.95	95.76	94.52	86.05	92.11
D&G	SDv1-5+DINOv2	73.26	66.56	67.51	65.31	66.46	93.30	94.25	92.05	93.20

Table 4: Comparison of absolute and relative robustness for DINO, Stable Diffusion, and Stable Diffusion+DINO methods under environmental conditions, including haze, snow, rain, dark, and strong sunlight.

Category	Method	Clean	Absolute Robustness						Relative Robustness					
			Haze	Snow	Rain	Dark	Strong Sunlight	Avg	Haze	Snow	Rain	Dark	Strong Sunlight	Avg
D	DINOv1	46.10	25.57	43.45	45.95	37.63	43.92	39.30	79.47	97.35	99.85	79.47	97.82	90.79
	DINOv2	63.17	56.42	62.57	63.32	59.24	62.86	60.88	93.25	99.40	100.15	93.25	99.69	97.15
G	SDv1-5	68.71	42.37	67.07	68.52	61.77	67.40	61.43	73.66	98.36	99.81	73.66	98.69	88.84
	DIFT->SDv2-1	72.84	58.54	72.51	72.61	69.43	72.59	69.14	85.70	99.67	99.77	85.70	99.75	94.12
D&G	SDv1-5+DINOv2	73.26	60.11	72.25	73.43	66.95	72.42	69.03	86.85	98.99	100.17	86.85	99.16	94.40